**MAHATMAGANDHIUNIVERSITY,KOTTAYAM**

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISSIONONWARDS)

# MG3DSEECT201-IntroductiontoMechatronics

**Duration:2hours** **MaximumMarks:70**

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S), Interest (I) and Appreciation (Ap)*

## Part A

MultipleChoiceQuestions Answer **All** Questions

Eachquestioncarries1mark

| | | | |
|---|---|---|---|
| 1 | Whichofthefollowingbestdescribesamechatronicsystem?<br>A) Apurelymechanicaldevice<br>B) Acombinationofelectronicsandsoftwareonly<br>C) Anintegratedsystemofmechanical,electronic,andcontrolelements<br>D) Asoftwaresimulationofamachine | U | 1 |
| 2 | Whichoneofthefollowingisanadvantageofusingmechatronicsinmanufacturing?<br>A) Increasesmanuallabor B)Reducesproduct quality<br>C)Enhancesprecisionand automation D)Decreasesuseoftechnology | U | 1 |
| 3 | InaSuccessiveApproximationRegistertypeADC,theconversionprocessis**:**<br>A) Serialandslow B)Basedonvoltagecomparisoninsteps<br>C)Non-digital D)Usedonlyinanalog computers | U | 1 |
| 4 | Whichofthefollowingcomponentsisusedtoconvertdigitalsignalstoanalogsignals?<br>A) SARADC B)MultiplexerC)R-2RladderDACD)SampleandHoldcircuit | U | 1 |
| 5 | Inthecontextofsensorsandtransducers,whichfactorismostimportantinsensor selection for mechatronics?<br>A) Sensorsize only B)Brandofthe sensor<br>B) Applicationrequirementssuchasaccuracy,range,andenvironment<br>C) Thecostliestoption | U | 1 |
| 6 | Whichofthefollowingscenarioswouldmostlikelyrequireaservomotoroverastepper motor?<br>A) Movinganobjectinfixed,discretestepsB)Continuousrotationwithlowaccuracy<br>B) Closed-looppositioncontrolwithfeedbackD)High-speedoperationwithoutfeedback | An | 2 |
| 7 | Whencomparingabeltdriveandageardrivein termsofmechanicalperformance, which statement is most accurate?<br>A) Beltdrivesofferbettertorquetransmissionthangeardrives<br>B) Geardriveshavehigherslippagecomparedtobelts<br>C) Beltdrivesarequieterbutlessefficientthangeardrives<br>D) Gearsarepreferredwhenlong-distancepowertransmissionisrequired | An | 2 |
| 8 | A mechatronic system requires precise position sensing in a linear motion application. Which sensor provides the most suitable performance**?**<br>A) Potentiometer B)Inductiveproximitysensor<br>C) Capacitivesensor D)LinearVariableDifferentialTransformer(LVDT) | An | 2 |
| 9 | Howdoesaninductiveproximitysensordifferfunctionallyfromacapacitiveproximity sensor when used in an automation setup?<br>A) Inductivesensorsdetectallmaterials,whilecapacitivesensorsdetectonlymetals<br>B) Capacitivesensorsrequiremagnetictargets<br>C) Inductive sensors detect metallic objects; capacitive sensors detect both metallic and non-metallic objects | An | 2 |

| | | | |
|---|---|---|---|
| | D) Capacitivesensorsarecontact-based;inductivesensorsare not | | |
| 10 | Inahydrauliccircuit,whatisthemostlikelyresultofamalfunctioningpressurecontrol valve?<br>A) Continuousincreaseinactuatorspeed<br>B) Failuretobuildnecessarypressure,leadingtosystemunderperformance<br>C) Increasedelectricalresistance<br>D) Airbubbleformationinthepneumaticsystem | An | 2 |
| 11 | Indigitalsignalprocessing,whatistheprimarydifferencebetweenanFIRfilter and an IIR filter?<br>A) FIRfiltershavefeedbackwhileIIRfiltersdonot<br>B) FIRfiltersarealwaysunstablewhileIIRfiltersarealwaysstable<br>C)FIR filters have finite impulse response whereas IIR filters have infinite impulse response<br>D) IIRfiltersarealwayslinearphasewhileFIRfiltersarenot | U | 3 |
| 12 | Indataacquisitionsystems,*sampling*refersto:<br>A)Convertingeachsampleintobinaryvalue<br>B) Holdingasignalconstantforashort interval<br>C) Multiplyingtwosignalstogether<br>D) Measuringacontinuoussignalatdiscretetimeintervals | U | 3 |
| 13 | Whichofthefollowingcommunicationprotocolsisamulti-master synchronousserialbus,commonlyusedforcommunicationbetweenICs onaPCB?<br>A) RS232 B) SPI C) $I^2C$ D)CAN | U | 3 |
| 14 | ZigBeeisawirelesscommunicationprotocolbestdescribedas:<br>A) Long-range,high-bandwidth,point-to-pointcommunication<br>B) Short-range,low-powermeshnetworkingforsensor/controlapplications<br>C) Medium-range,satellite-basedcommunication<br>D)Ultra-widebandcommunicationforvideostreaming | U | 3 |
| 15 | Insignalconditioning,*pre-amplification*isusedmainlyto:<br>A) Digitallyfilterthesignal<br>B) Convertadigitalsignalintoanalog<br>C)Increaselow-levelsensorsignalsbeforefurtherprocessing<br>D) RemovenoisefromACpowerlines | U | 3 |
| 16 | Whichembeddedlanguageiscommonlyusedforprogrammingmicrocontrollersin mechatronics applications?<br>A) Java B) HTML<br>B) EmbeddedC D)MATLAB | A | 4 |
| 17 | Whatdoesapotentiometermeasurewhenconnectedtoan Arduino<br>A) Lightintensity B)Voltageorposition<br>C)Temperature D)Currentfrequency | A | 4 |
| 18 | Inaproximity sensor application,which type of proximitysensor is bestsuitedfor detecting metallic objects<br>A) Ultrasonicsensor B)CapacitivesensorC)Inductive sensor D)Lightsensor | A | 4 |
| 19 | Whatdoesreal-timemonitoringmeaninamechatronicssystem<br>A) Controlhappenslater B)Systemworkswithoutfeedback<br>C)Thesystemischeckedandcontrolledatthesame time D)Onlydatais saved | A | 4 |
| 20 | Whichcontrolmethodhelpskeeparoboticarmatthecorrectposition<br>A) Open-loop B)Manual onlyC)Relayswitch D)PID control | A | 4 |

**1 * 20 = 20**

## Part B

Short Answer Questions

Answer any **6** QuestionsEachquestioncarries5marks

| | | | |
|---|---|---|---|
| 21 | Whatisthemaindifferencebetweenaconventionalsystemandamechatronic system? | U | 1 |
| 22 | BrieflyexplaintheworkingprincipleofanR-2RladderDigitaltoAnalog Converter | U | 1 |
| 23 | Compareandcontrastthefunctionsofarelayandasolenoidinamechanical control system. | An | 2 |
| 24 | WhyisanLVDTconsideredmoreaccuratefordisplacementmeasurement compared to a potentiometer? | An | 2 |
| 25 | Explainthepurposeof*sampling*and*quantization*indigitalsignal processing. | U | 3 |
| 26 | Whatis***CAN***networkingandwhereisitcommonlyused? | U | 3 |
| 27 | Describetheprocessofinterfacingapotentiometerandaproximitysensorwith an Arduino. Include a simple block diagram or circuit example. | A | 4 |
| 28 | Explainhowaservomotorandpotentiometerareusedtogethertocontrola simple robotic arm. | A | 4 |

**6 * 5 = 30**

## Part C

Essay

QuestionsAnswerany**2**Q uestions

Eachquestioncarries10marks

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|---|---|---|---|
| 29 | Explain the role and importance of Mechatronics in modern manufacturing and industrial automation. Include examples of real-world applications and describe how sensors and hardware components contribute to system efficiency. | U | 1 |
| 30 | Discuss the selection and integration of mechanical actuators and sensors in mechatronic systems. Analyze servo motors, linear actuators, and proximity sensors? | An | 2 |
| 31 | With neat block diagrams, explain the working of a closed-loop control system. Highlight the role of sensors, controllers, actuators, and feedback. Give two practical examples. | U | 3 |
| 32 | Describeareal-worldcasestudyofanindustrialrobotusedinmanufacturing. Explainitsworking,themainhardwareandsensorsusedandhowautomation improves productivity and precision. | A | 4 |

**2 * 10 = 20**